

Stable Normalizer Untwisting and the Finite-Corner Obstruction

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Abstract

We analyze the reduced alignment problem for a Cartan subalgebra $D \subset A$, where A is the hyperfinite II_1 factor. A cocycle action (α, Ω) preserves D , its two-cocycle takes values in $\mathcal{N}_A(D)$, and it is exterior equivalent through A -unitaries to a prescribed genuine Bernoulli action. We prove that after passing to $A \overline{\otimes} \mathcal{B}(\ell^2 G)$, the cocycle is a coboundary through normalizers of $D \overline{\otimes} \ell^\infty(G)$. We then show that finite normalizer untwisting is equivalent to a rank-one corner problem for the stabilized genuine action.

1 The reduced cocycle action

Let G be countable, let $\sigma : G \curvearrowright A$ be a genuine trace-preserving action, and let $D \subset A$ be Cartan. Assume there are $a_g \in \mathcal{U}(A)$ such that

$$\alpha_g = \text{Ad}(a_g) \circ \sigma_g, \quad \alpha_g(D) = D.$$

Set

$$\Omega(g, h) = a_g \sigma_g(a_h) a_{gh}^* \in \mathcal{N}_A(D).$$

Then

$$\alpha_g \alpha_h = \text{Ad}(\Omega(g, h)) \alpha_{gh}.$$

The finite normalizer-untwisting problem is to find $n_g \in \mathcal{N}_A(D)$ satisfying

$$n_g \alpha_g(n_h) \Omega(g, h) = n_{gh}. \tag{1}$$

2 Exact stable normalizer untwisting

Put

$$\tilde{A} = A \overline{\otimes} \mathcal{B}(\ell^2 G), \quad \tilde{D} = D \overline{\otimes} \ell^\infty(G),$$

and let $(e_{r,s})$ be the standard matrix units. Define

$$\beta_g = \alpha_g \otimes \text{Ad}(\lambda_g)$$

and

$$W_g = \sum_{r \in G} \Omega(g, g^{-1}r) \otimes e_{r,r}.$$

Theorem 2.1 (Stable normalizer untwisting). *For every $g \in G$,*

$$W_g \in \mathcal{N}_{\tilde{A}}(\tilde{D})$$

and

$$\beta_g(W_h) W_g = (\Omega(g, h) \otimes 1) W_{gh}.$$

Consequently,

$$\gamma_g = \text{Ad}(W_g^*) \circ \beta_g$$

is a genuine action preserving $\tilde{D}$.

Proof. The diagonal coefficients of W_g normalize D , so W_g normalizes $\tilde{D}$. At the diagonal coordinate r , the left-hand side is

$$\alpha_g(\Omega(h, h^{-1}g^{-1}r)) \Omega(g, g^{-1}r).$$

The cocycle identity

$$\alpha_g(\Omega(h, k)) \Omega(g, hk) = \Omega(g, h) \Omega(gh, k)$$

with $k = h^{-1}g^{-1}r$ gives the claimed identity. Taking adjoints shows that the cocycle defect cancels in $\gamma_g \gamma_h$. $\square$

3 The rank-one corner

Let

$$p = 1 \otimes e_{e,e}, \quad p_g = 1 \otimes e_{g,g}, \quad r_g = 1 \otimes e_{e,g}.$$

Then

$$\gamma_g(p) = p_g, \quad \gamma_g(x \otimes e_{e,e}) = \alpha_g(x) \otimes e_{g,g},$$

and

$$r_g \gamma_g(r_h) = (\Omega(g, h) \otimes e_{e,e}) r_{gh}.$$

Proposition 3.1 (Finite-corner criterion). *The following are equivalent.*

1. There are $n_g \in \mathcal{N}_A(D)$ satisfying (1).
2. There are $s_g \in \mathcal{GN}_{\tilde{A}}(\tilde{D})$ such that

$$s_g^* s_g = p_g, \quad s_g s_g^* = p,$$

and

$$s_g \gamma_g(s_h) = s_{gh}.$$

One may take $s_g = n_g \otimes e_{e,g}$.

Proof. For $s_g = n_g \otimes e_{e,g}$,

$$\gamma_g(s_h) = \alpha_g(n_h) \Omega(g, h) \otimes e_{g,gh}.$$

Hence

$$s_g \gamma_g(s_h) = n_g \alpha_g(n_h) \Omega(g, h) \otimes e_{e,gh},$$

which equals s_{gh} exactly when (1) holds. Conversely, every groupoid normalizer carrying $p_g \tilde{D} p_g$ onto $p \tilde{D} p$ has this form. $\square$

4 A smaller target: one nonzero coherent graph corner

The special alignment problem contains additional information. Define

$$t_g = a_g^* \otimes e_{e,g}.$$

Then

$$t_g^* t_g = p_g, \quad t_g t_g^* = p, \quad t_g \gamma_g(t_h) = t_{gh}.$$

Thus a coherent return system already exists in $\tilde{A}$, although it need not normalize $\tilde{D}$. It is enough to find a nonzero normalizing subsystem.

Identify A with $p \tilde{A} p$ and D with $p \tilde{D} p$.

Proposition 4.1 (Nonzero-corner reduction). *Let $0 \neq q \in D$. Suppose that there are $s_g \in \mathcal{GN}_{\tilde{A}}(\tilde{D})$ satisfying*

$$s_g^* s_g = \gamma_g(q), \quad s_g s_g^* = q, \quad s_g \gamma_g(s_h) = s_{gh}.$$

Then

$$w_g = s_g t_g^* \in A$$

satisfies

$$w_g w_g^* = q, \quad w_g^* w_g = \sigma_g(q), \quad w_g \sigma_g(w_h) = w_{gh}, \quad (2)$$

and

$$n_g := w_g a_g^* \in \mathcal{GN}_A(D).$$

Conversely, every family with these properties gives such a coherent normalizing subsystem by $s_g = w_g t_g$.

Proof. The compression identity

$$t_g \gamma_g(x) t_g^* = \sigma_g(x), \quad x \in p \tilde{A} p,$$

and the coherence of $(t_g)_g$ give (2) by direct multiplication. Since both s_g and the matrix unit in t_g^* have the indicated source and range, $n_g = w_g a_g^*$ has source $\alpha_g(q)$, range q , and normalizes the corresponding corners of D . The converse follows from the same identities in reverse. $\square$

The family $(w_g)_g$ in (2) is a generalized σ -cocycle of support $\tau(q)$. For the Krogager–Vaes action, a proper support is impossible.

Here the Krogager–Vaes action means

$$A = \overline{\bigotimes_{k \in \Gamma} (A_0, \tau_0)}, \quad \theta : \Gamma \curvearrowright (A_0, \tau_0),$$

where (A_0, τ_0) is a nontrivial amenable finite tracial algebra and θ is trace preserving, with

$$\sigma_{(g,h)}(\pi_k(x)) = \pi_{gkh^{-1}}(\theta_h(x)), \quad g, h, k \in \Gamma, x \in A_0. \quad (3)$$

Here $\pi_k : A_0 \rightarrow A$ denotes the k -th tensor-leg embedding.

Proposition 4.2 (Proper-support rigidity). *Let $G = \Gamma \times \Gamma$, with Γ nonamenable, and let σ be the Krogager–Vaes left–right Bernoulli action. If $(w_g)_g$ satisfies (2) for a nonzero projection $q \in A$, then $q = 1$.*

Proof. Repeat the nonamenable-centralizer part of the proof of [2, Theorem 3.2], with the arbitrary-base extension in [2, Remark 3.5]. Take

$$H = \{e\} \times \Gamma, \quad H' = \Gamma \times \{e\} \subset C_G(H).$$

The restriction to H' is the ordinary noncommutative Bernoulli shift on the coordinate set Γ ; the coordinate stabilizers are trivial, so it has stable spectral gap. The restriction to H is weakly mixing: on finite-support tensors, translate one coordinate support away from the other, and observe that the trace-preserving internal automorphism θ_h does not affect the orthogonality calculation. Finally, extend θ_h to $A_0 * L(\mathbb{Z})$ by the identity on the free $L(\mathbb{Z})$ -leg. The standard free malleable deformation then commutes with the full left–right action.

These are exactly the spectral-gap, weak-mixing, and malleability inputs in Popa’s support argument. Applied to the exact generalized cocycle $(w_g)_g$, that argument forces $\tau(q) = 1$. Therefore $q = 1$. $\square$

Corollary 4.3 (A nonzero graph corner gives exact alignment). *For the Krogager–Vaes action, the existence of the subsystem in Proposition 4.1 implies that (1) has a solution in $\mathcal{N}_A(D)$ and that an inner conjugate of D is exactly σ -invariant.*

Proof. Proposition 4.2 gives $q = 1$. Thus the $n_g = w_g a_g^*$ are unitaries in $\mathcal{N}_A(D)$, and the cocycle identity in (2) is equivalent to (1). Also $w : G \rightarrow \mathcal{U}(A)$ is now an ordinary σ -cocycle. Noncommutative cocycle superrigidity [3, Theorem 4.1] gives $u \in \mathcal{U}(A)$ and a character $\chi : G \rightarrow \mathbb{T}$ such that

$$u^* w_g \sigma_g(u) = \chi(g).$$

With $z = u^*$,

$$z^* \sigma_g(z) = \chi(g)^* w_g = (\chi(g)^* n_g) a_g.$$

Since $\chi(g)^* n_g \in \mathcal{N}_A(D)$, the Cartan zDz^* is σ -invariant. $\square$

5 The correct graph-projection formulation

Let

$$Z_{\tilde{D}} = \tilde{D}' \cap \langle \tilde{A}, e_{\tilde{D}} \rangle.$$

Write $\tilde{\tau}$ for the canonical semifinite trace on $\tilde{A}$, and Tr for the canonical trace on the basic construction. This is the abelian relation algebra of the stabilized Cartan pair.

Proposition 5.1 (Graph-projection criterion). *Let $v \in \tilde{A}$ be a partial isometry with $v^*v, vv^* \in \tilde{D}$ of finite trace. Then*

$$v \in \mathcal{GN}_{\tilde{A}}(\tilde{D}) \iff ve_{\tilde{D}}v^* \in Z_{\tilde{D}}.$$

For a unitary v , the equivalent standard-form identity is

$$(vJvJ)e_{\tilde{D}}(vJvJ)^* = e_{\tilde{D}}.$$

Here J is the modular conjugation on $L^2(\tilde{A})$. In general, left multiplication by a normalizer does not fix $e_{\tilde{D}}$.

Proof. If v is a groupoid normalizer, a direct computation shows that $ve_{\tilde{D}}v^*$ commutes with $\tilde{D}$. Conversely, commutation implies

$$(v^*dv)e_{\tilde{D}} = e_{\tilde{D}}(v^*dv)$$

on the source of v , for every $d \in \tilde{D}$. Applying this identity to the finite-trace source vector $v^*v \in L^2(\tilde{A})$ shows that v^*dv belongs to the corresponding corner of $\tilde{D}$. Writing $p = v^*v$ and $q = vv^*$, the algebra $v^*\tilde{D}qv$ is a maximal abelian subalgebra of $p\tilde{A}p$ contained in $\tilde{D}p$, so the two are equal. Equivalently, $v\tilde{D}pv^* = \tilde{D}q$. The standard-form identity implements $\text{Ad } v$ simultaneously on the left and right. $\square$

Thus the deformation problem is an atomicity problem: extract a coherent family of graph projections in $Z_{\tilde{D}}$.

Conjecture 5.2 (Nonzero coherent graph extraction). *For the Krogager–Vaes action, the stabilized coherent system $(t_g)_g$ has a nonzero coherent subsystem $(s_g)_g$ as in Proposition 4.1.*

One possible quantitative detector is

$$\text{pur}_{\tilde{D}}(v) = \left\| E_{Z_{\tilde{D}}}(ve_{\tilde{D}}v^*) \right\|_{2, \text{Tr}}^2.$$

For a nonzero finite-trace partial isometry v , faithfulness and the conditional-expectation contraction give

$$0 < \text{pur}_{\tilde{D}}(v) \leq \text{Tr}(ve_{\tilde{D}}v^*) = \tilde{\tau}(vv^*).$$

It is useful to minimize the complementary graph defect

$$\delta_{\tilde{D}}(v) = \tilde{\tau}(vv^*) - \text{pur}_{\tilde{D}}(v) = \left\| ve_{\tilde{D}}v^* - E_{Z_{\tilde{D}}}(ve_{\tilde{D}}v^*) \right\|_{2, \text{Tr}}^2.$$

By Proposition 5.1, $\delta_{\tilde{D}}(v) = 0$ exactly when v is a graph normalizer. A deformation proof should drive this defect to zero in four stages.

1. Approximate the hyperfinite Cartan relation by finite subrelations and construct almost coherent vectors carrying the full Feldman–Moore phase data.
2. Use the stable spectral gap of one free factor to prevent these vectors from escaping weakly to zero under the Bernoulli malleable deformation.
3. Use weak mixing of the commuting factor to eliminate the diffuse part of the limiting relation kernel and retain a nonzero graph component.
4. Take fiberwise polar parts. This gives the subsystem in Proposition 4.1; Proposition 4.2, rather than an ergodicity or Zorn argument, upgrades it to full support.

6 Conclusion

Stabilization provides a tautological coherent orbit bundle, but it does not by itself descend that bundle to the finite corner. The remaining new statement is now narrower: Bernoulli spectral gap must extract one nonzero coherent graph component from the stabilized bundle. Proper-support rigidity then makes the component full, and noncommutative H^1 -superrigidity produces exact invariant alignment.

References

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